

# Rewritten Chain of Thought for the Solution to the Two-Minus Water-Wave Amplitude

## Abstract

This document is a first-person reconstruction of the chain of thought for the original AI derivation of the closed-form tree amplitude  $A_n$  in the two-minus sector of one-dimensional deep-water surface waves. The run reached a verified all-kinematics closed form: a truncated-power / inclusion–exclusion “spline” over subset-sum chambers, checked against the supplied **BGAmplitude** to exact rational equality (relative error identically 0) for  $n = 4, \dots, 7$  in many chambers and at  $g = 1, 2, 3$ . This reconstruction is rebuilt from the visible run trajectory — the raw chain of thought was encrypted, so only the visible narration, tool calls, and outputs survive — and it keeps the original wording, the false starts, and the hedges (the **Indeterminate** at  $n = 4$ , the protected-**List** loop errors, the wall of garbage symbolic output) rather than tidying them away. I also note where the prompt’s hint did real work and where I leaned on it perhaps harder than I needed to.

## Rewritten Chain of Thought

I am allowed to read only `prompt.md` and `OnShellBG.m`, and I work only from those plus data I generate myself; the deliverable goes in `fugu.ultra`. The prompt is unusually generous: it tells me up front that the answer is a *piecewise homogeneous polynomial* in the  $\omega_i$  — a genuine polynomial on each chamber, with the two-minus kinematic space cut into chambers, and a different homogeneous polynomial on each. That is a strong hint and I take it at face value from the start: my job is not “is it piecewise?” but “what are the chamber walls and what polynomial sits on each piece.” Reading `OnShellBG.m`, I see immediately where the pieces must come from: every kernel and every propagator is built out of  $\text{mag}[k] = |k|$ , and on this sector  $k_i = \sigma_i \omega_i^2 / g$ , so the only non-analytic objects in the whole construction are absolute values of signed sums  $\sum_i \sigma_i \omega_i^2$ . An  $|\cdot|$  has a corner where its argument changes sign; that is exactly a chamber wall. So the hint and the code agree before I run anything: the walls will be subset-sum surfaces.

First, is `wolframscript` available? It is. I run the file as-is, and that is mildly wasteful: `OnShellBG.m` ends in a **VI. TESTS** block that demonstrates a *different* sector,  $\sigma = (-1, +1, \dots, +1)$ , where the amplitude is identically zero, so it prints “ $A_5 = 0, A_6 = 0, A_7 = 0$ ” and then grinds on  $n = 8$ . I let it run, then stop leaning on it. Better: split the file as text at the **VI. TESTS** marker with `awk` and keep only the definitions in a `BGdefs.generated.m`, so I load the recursion without the demo. (Small frictions follow all the way through: a `Do[...,{n,fw},...]` trips “Tag List is Protected,” so I switch those loops to `Scan`; later a `Count[... , s_/; ...]` with a bound pattern throws a syntax error and I rewrite it as `Length@Select`.)

Now the oracle behaves, and the first real surprise is  $n = 4$ . Direct rational evaluation returns **Indeterminate** — the messages are “1/0” (`Power::infy`) and “0 · ComplexInfinity” (`Infinity::indet`). The reason is structural and I pin it down later: at  $n = 4$  the solver always returns  $\omega = (-b, a, b, -a)$ , so legs 2 and 4 carry exactly opposite momenta,  $k_2 + k_4 = -a^2 + a^2 = 0$ , and the BG current on  $\{2, 4\}$  hits a propagator  $-i/(\omega_S^2/|k_S| - g)$  with  $|k_S| = 0$  — a 1/0 multiplied by a vanishing numerator, a removable 0/0. So I set  $n = 4$  aside, to recover as a limit, and look at  $n = 5, 6, 7$  with generic positive free frequencies. Those are finite and purely imaginary rationals: e.g. for free  $\{2, 3, 4\}$  I get  $A_5 = -\frac{8704}{3}i$ , for  $\{4, 3, 2\}$  I get  $A_5 = -19968i$ , for  $\{1, 2, 10\}$  I get  $-\frac{2176}{13}i$ . Writing  $A_n = iP_n$  and chasing  $P_n$  is the obvious move.

For  $n = 4$  I do not have to guess the chamber polynomial — I can ask Wolfram directly. I feed symbolic free frequencies  $\{a, b\}$ , get the symbolic **BGAmplitude** (a genuinely ugly thing, full of nested  $\text{Abs}[a^2 - b^2]$ ,  $\text{Abs}[a^2 + b^2]$ , and the like), and **FullSimplify** it under sign assumptions on the two chambers:

$$b > a : \quad A_4 = -8ia^3b, \quad a > b : \quad A_4 = -8iab^3.$$

Running all eight sign-cells of  $(a, b)$  collapses everything to just these two polynomials,  $-8ia^3b$  and  $-8iab^3$ , i.e.  $A_4 = -8iab \min(a^2, b^2)$ . That is precisely a two-piece spline with a wall at  $a^2 = b^2$ , exactly the kind of object the hint promised, and it gives me the overall normalization  $i 2^{n-1} \omega_1 \omega_2$  ( $2^{n-1} = 8$  at  $n = 4$ ) to factor out.

Now  $n = 5$ , where the chamber structure becomes informative. I do *not* trust `FullSimplify` to crack the symbolic five-point amplitude — it stalls for thirty seconds and I interrupt it. Instead I resolve the absolute values by hand, chamber by chamber: collect every `Abs` argument in the symbolic amplitude (there are 45 of them, things like  $a^2 - b^2$ ,  $a^2 - b^2 - c^2$ ,  $b^2 + c^2$ , and worse), replace each `Abs[x]` by `Sign[x] · x` evaluated *at a sample point inside the chamber*, and then `Together/Cancel/Factor`. Inside each chamber this turns the amplitude into an honest rational function. At the sample  $(a, b, c) = (2, 3, 4)$  (the smaller negative-leg square below all positive squares) I get

$$A_5 = \frac{-16 i a^5 (ab + b^2 + ac + bc + c^2)}{a + b + c},$$

which numerically is  $-\frac{8704}{3}i$  — matching the oracle. At  $(4, 3, 2)$  a different chamber gives  $-32 i a b^2 c^2 (ab + b^2 + ac + bc + c^2)/(a + b + c)$ , again matching  $-19968 i$ . These are not unrelated answers; the leading  $a^5$  in the first and the  $ab^2c^2$  in the second are the two ends of a running inclusion–exclusion. Combined with the  $n = 4$  result they tell me what the chamber polynomial is.

Here is the structural reading. Write  $q_i = \omega_i^2$ , let the two  $\sigma = -1$  legs be  $1, 2$  and the positive legs be  $P = \{3, \dots, n\}$ . The walls are the subset-sum surfaces  $q_S = q_1$  for  $S \subseteq P$  (equivalently  $q_{P \setminus S} = q_2$ , since  $\sum_{a \in P} q_a = q_1 + q_2$  on shell). In the soft chamber where  $q_1$  is below every  $q_S$ , only the empty subset survives and the polynomial is the bare monomial  $q_1^{n-3}$  — which, restored with the prefactor, is  $A_n = i 2^{n-1} \omega_1 \omega_2 \omega_1^{2n-6}$ , i.e. the leading  $a^5$  piece at  $n = 5$ . As  $q_1$  crosses each wall, an inclusion–exclusion correction switches on. The whole object is one truncated power. With the clamped power  $[x]_+^m = \max(x, 0)^m$ ,

$$A_n = \frac{i 2^{n-1}}{g^{n-3}} \omega_1 \omega_2 \sum_{S \subseteq \{3, \dots, n\}} (-1)^{|S|} [q_1 - \sum_{a \in S} q_a]_+^{n-3}$$

and on a fixed open chamber  $C$ , where the active set  $\mathcal{I}_C = \{S : q_S < q_1\}$  is constant, this is literally a homogeneous polynomial of total frequency degree  $2n - 4$ ,

$$A_n|_C = \frac{i 2^{n-1}}{g^{n-3}} \omega_1 \omega_2 \sum_{S \in \mathcal{I}_C} (-1)^{|S|} (q_1 - q_S)^{n-3},$$

which is exactly the “piecewise homogeneous polynomial” the hint described. The  $g$ -power is fixed because every  $|k| = \omega^2/g$ , and the homogeneity degree  $2n - 4$  matches the scaling I expected.

I want to be candid that the hint did real work here. It told me ahead of time that the answer is polynomial on each piece, so when my chamber-by-chamber simplifications came back as clean rationals with a common  $(a + b + c)$  denominator, I trusted that the denominator was an artifact of the resonant parametrization and that the underlying chamber object was a polynomial — I did not waste time chasing a rational-function ansatz. The hint also pre-named the structure (“chambers”), so the moment I saw the subset-sum `Abs` arguments I matched them to walls without hesitation. If anything I under-tested the *converse*: I never stress-tested whether some chamber could secretly carry a non-polynomial piece, because the prompt had already promised me it could not.

Two things remain. For  $n = 4$  the formula has exponent  $m = n - 3 = 1$ , so it predicts  $A_4 = 8i \omega_1 \omega_2 [q_1 - (q_1 - q_3)_+ - (q_1 - q_4)_+] = -8iab \min(a^2, b^2)$  — exactly the removable-singularity limit I extracted symbolically, even though a *direct* rational  $n = 4$  evaluation still returns `Indeterminate` (this is a property of the supplied code at the degenerate  $k_2 + k_4 = 0$  point, not of the formula). I report the  $n = 4$  rows via that symbolic-chamber limit: free  $\{2, 3\}$  and  $\{3, 2\}$  both give  $-192i$ , and  $\{-2, 5\}$  gives  $320i$ , all matched.

Then I verify. I implement the truncated-power formula from scratch (`ppow[x,m]:=If[x>0,x^m,0]`, a subset sum over `Subsets[Range[3,n]]`) and compare `Simplify[BGAmplitude]` against it in exact rational arithmetic. The difference is identically zero — not “small,” but `Simplify[BG – formula] = 0` before any

floating-point step — for  $n = 5, 6, 7$  at many points, and I deliberately spread the points across chambers by counting how many subsets are active. The tests hit, for example,  $4/8, 6/8, 7/8$  active subsets at  $n = 5$ ;  $8/16, 10/16, 12/16, 15/16$  at  $n = 6$ ; and  $16/32, 24/32, 30/32, 31/32$  at  $n = 7$  — so I am not sitting in one cell. I also confirm the gravity scaling separately: at free  $\{2, 3, 4\}$ ,  $n = 5$ , the BG values at  $g = 1, 2, 3$  ( $-\frac{8704}{3}i, -\frac{2176}{3}i, -\frac{8704}{27}i$ ) track the formula's  $g^{-(n-3)}$  exactly, ratio 1 in each case.

So, honestly: the run succeeded. Within the prompt's own kinematics — and, as the chamber sweep shows, well beyond a single chamber — the boxed truncated-power formula reproduces **BGAmplitude** to exact rational equality for  $n = 4, \dots, 7$ , with the chamber decomposition stated explicitly and the  $n = 4$  removable singularity resolved as a limit. The one structural fact I leaned on rather than independently re-derived is the polynomiality-per-chamber that the hint handed me; everything downstream of that — the walls  $q_S = q_1$ , the alternating-sign inclusion–exclusion, the  $i 2^{n-1} \omega_1 \omega_2$  prefactor, the  $n = 4$  limit — I forced out of the oracle's own data and checked against it.